

Scalable Research Infrastructure in a Business School Context: A Case Study of the Data Science Research Services

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Abstract—This paper presents a detailed overview of the Data Science Research Services (DSRS) at the Gies College of Business, at the University of Illinois, an integrated science gateway that serves as a critical component of the college’s research infrastructure. The DSRS unit, supported by a set of computational resources and innovative cloud technologies, facilitates data-intensive research through a user-friendly interface, supporting a wide variety of applications ranging from machine learning models to advanced data visualization tools. This infrastructure provides an efficient collaborative environment for students, faculty, and researchers alike, fueling data-driven research within the Business School and other units on Campus. This paper explores the technical and architectural aspects of the DSRS, its role in enabling scalable research, and the strategies employed for data management, scalability, and ensuring the reproducibility of research in a business school. Furthermore, the paper sheds light on the user interaction model within the DSRS, emphasizing its role in fostering a collaborative research ecosystem. This work provides valuable insights for other academic institutions seeking to build or enhance their research infrastructures, particularly within the business, marketing, and finance disciplines.

I. INTRODUCTION

The advent of big data and sophisticated computational technologies has ushered in a new era of research possibilities across various disciplines. As the business world becomes more data-driven, the need for high-performance, scalable, and collaborative research infrastructure in business schools has never been more pertinent. This paper presents a case study of the Data Science Research Services (DSRS)[1] at the Gies College of Business at the University of Illinois, an example of a robust, scalable, and versatile science gateway that sits at the intersection of business academia and cutting-edge technology.

DSRS was conceived within the Disruption and Innovation unit at the college with the vision to equip faculty, PhD students, and researchers with the necessary computational tools, methodologies, data science techniques, and resources to drive data-centric research. Through a wide array of services and state-of-the-art cloud technologies, DSRS facilitates an efficient, user-friendly, and collaborative research environment that supports various applications, from machine learning and data science to advanced data visualization tools.

In this paper, we delve into the detailed architecture and technical components that constitute the DSRS platform. It presents the intricate interplay between various cloud technologies, computational resources, and data management strategies, contributing to the scalability and reproducibility of research within the Gies College of Business. Furthermore, we shed light on DSRS’s role in fostering a collaborative and innovative research ecosystem and the lessons learned from this endeavor. This case study serves as a valuable resource for other academic institutions aiming to enhance their research infrastructures, especially within the domain of business and management.

II. BUILDING THE DSRS INFRASTRUCTURE

Creating an effective and efficient research infrastructure, particularly for data-intensive research, requires careful planning and strategic execution. In the case of DSRS at the Gies College of Business, we had to cater to a diverse set of needs, ranging from machine learning model deployment, web scraping, text analysis, and advanced data visualization among others. In order to meet these needs, we capitalized on a rich combination of computational resources and innovative cloud technologies.

The DSRS infrastructure’s backbone is a high-performance computing cluster consisting of over 640 cores, 2.5 TB of RAM, and over 30TB of shared storage, in addition to 2.8 TB of SSD storage. This set up provides substantial computational power to support a multitude of research projects simultaneously.

The cluster, managed by Openstack [2], is utilized to provide virtual machines for individual applications, including our blockchain applications and other environments tailored to specific research groups. Within this architecture, we also deploy a Kubernetes[3] cluster using Terraform[4] and Rancher[5], both cloud technologies that offer flexibility and scalability.

On top of this Kubernetes cluster, we deploy various applications managed by Argo CD[6], a continuous integration

platform. This platform helps maintain the stability and reliability of our services, thanks to its version control and preservation of the cluster status after every change. All of our manifests and configurations are on Github[7] which is directly connected and in sync with Argo.

An important part of the infrastructure is our Jupyter Hub [8] environment, used both internally and by other research groups within the college. We have the capability to provide a hub to a group or to deploy individual labs for particular needs with a line of configuration. This environment, complemented by frameworks like Dask[9], Spark[10], H2O.ai[11] and, funcx[12] that supports the parallel execution of machine learning models and makes interactive, data-driven research readily accessible.

To ensure our infrastructure supports a wide variety of research activities, we've incorporated different databases such as MongoDB[13], PostgreSQL[14], and MySQL[15]. Moreover, we have a dropbox-like infrastructure for seamless file upload, download, and sharing. The security of our platform is a high priority, with Dex[16] handling authentication and authorization.

III. ENABLING RESEARCH: THE DSRS APPLICATIONS

The DSRS infrastructure, while technologically advanced, serves a singular purpose: to enable and amplify data-driven research within the Gies College of Business. The suite of applications deployed within our Kubernetes cluster has been carefully chosen and custom-built to empower researchers and make complex data science methods accessible and straightforward.

As described, one of the essential components of our application portfolio is Jupyter Hub. The Jupyter notebook, with its interactive computing environment, allows users to create and share documents that contain live code, equations, visualizations, and narrative text. With this, researchers can conveniently conduct end-to-end data science projects, from data cleaning and transformation to statistical modeling and data visualization.

Further supplementing our data analysis capabilities are machine learning frameworks such as H2O.ai, Kubeflow, and many others. These platforms allow us to leverage the full computational power of our infrastructure, running complex machine learning models in parallel and at scale. This power broadens the range of research questions that can be explored and the speed with which insights can be derived.

To cater to the diverse research needs, we've enabled the deployment of dashboards using Tableau[17] and Python web interfaces like Streamlit[18]. These tools help researchers distill complex data into understandable visual narratives, a crucial aspect of modern research communication.

Our infrastructure also supports text and data mining, allowing researchers to extract valuable insights from large, unstructured datasets. With web scraping capabilities, DSRS can fetch data from the internet, expanding the range of available data sources for research.

For collaborative work, we provide a markdown app using Hedgedoc[19] that allows users to write markdown files collaboratively. Additionally, we utilize OpenFaaS[20], a Function as a Service platform, to deploy any function quickly, facilitating rapid prototyping and experimentation. This also allows one to carry out many repeated tasks at once. Such tasks include converting a PDF file into text, extracting information from images, generating unique identifiers, and collecting all data from a given company.

Figure 1 shows a schematic diagram of our current infrastructure with some of the applications used by the DSRS team.

Moreover, to uphold the principles of reproducible research, we maintain a private internal registry for Docker images and manage all configuration files on GitHub. This practice allows us to keep track of changes, reproduce previous states, and troubleshoot effectively.

IV. DATA MANAGEMENT AND SCALABILITY

In the world of data science, handling large volumes of data efficiently is a critical aspect. The DSRS infrastructure is designed to address this challenge, offering robust data management capabilities and excellent scalability.

The foundation of our data management strategy is a mix of databases including MongoDB, PostgreSQL, and MySQL. These databases, each with its distinct advantages, offer versatile and robust solutions for storing and retrieving data the college has purchased or has access to. They cater to the diverse needs of various research projects, whether they require document-oriented databases, object-relational databases, or more traditional relational databases.

Data storage is another crucial aspect of our data management. The DSRS infrastructure includes over 30TB of shared storage and 2.8TB of SSD storage managed by Cinder within Openstack. This vast amount of storage is designed to accommodate large data sets, essential in many contemporary research scenarios. We provide many mechanisms to access multiple sources of research-related data by interacting with multiple APIs.

Beyond simply storing data, we have made strides in managing and organizing data more effectively. Our Dropbox-like infrastructure, for example, allows users to upload, download, and share files easily. We can configure multiple boxes with varying permissions, including read-only and read-write boxes, enhancing data accessibility while maintaining control and security.

Scalability is a fundamental aspect of our infrastructure. With over 640 cores and 2.5TB of RAM, we have a significant amount of computational resources. However, the true power of our infrastructure lies in its design, which allows these resources to be dynamically allocated and scaled based on the requirements of each project. This elasticity, made possible by OpenStack and Kubernetes, ensures that every research project can leverage the amount of computational power it needs, without wasting resources or slowing down other projects. In addition, through research awards, we have access to cloud computing credits from Microsoft Azure which allows us

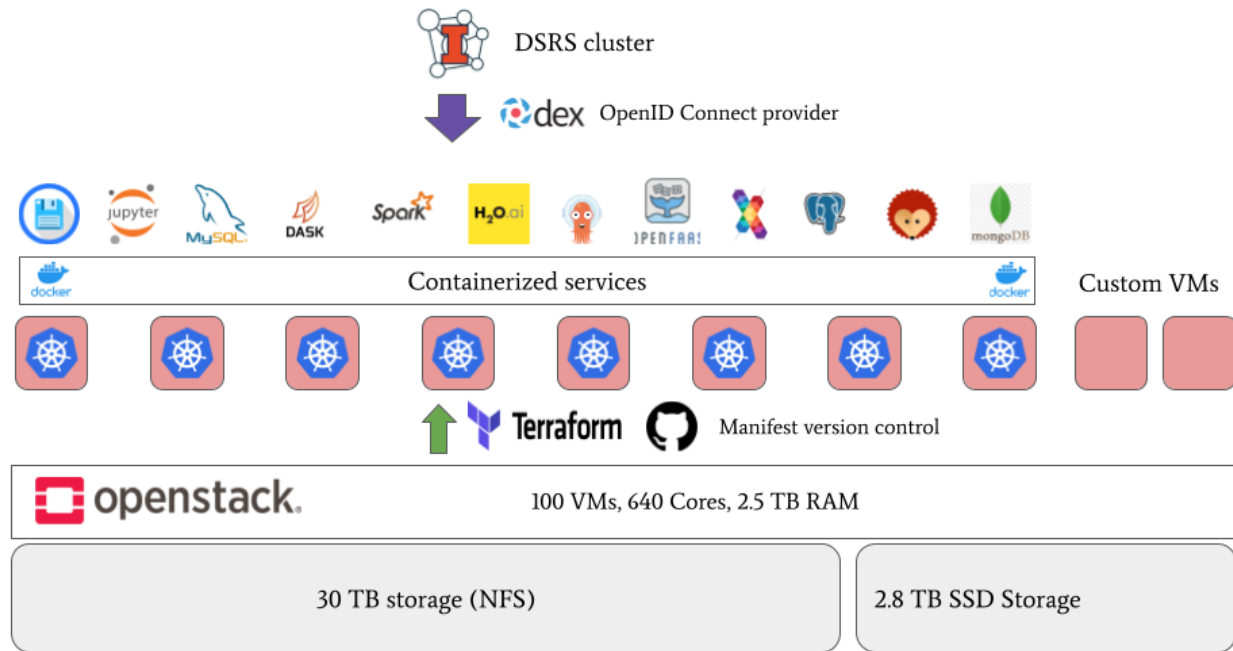


Fig. 1. DSRS Infrastructure - An overview of the integrated technologies supporting data science research at Gies College of Business

to expand our capabilities beyond what our current cluster can offer, including GPU access, Windows server, and other services from the marketplace.

Our data management and scalability strategies serve as the backbone of our infrastructure, allowing us to handle vast amounts of data efficiently and dynamically allocate computational resources. These capabilities are crucial in the era of big data and high-performance computing, significantly enhancing our capacity to conduct advanced data science research for the college.

V. USER INTERACTION AND COLLABORATION

At DSRS, our infrastructure is not only designed for handling data and computations but is also structured to facilitate seamless user interactions and collaborations. Central to our philosophy is the belief that scientific progress is best achieved in a conducive collaborative environment, and our infrastructure reflects this.

Our user collaboration extends to the development of our infrastructure itself. All our configurations are managed via GitHub, offering version control and the ability to track and roll back changes if necessary. By open-sourcing our configuration files, we promote transparency and allow the collective wisdom of the community to enhance our infrastructure.

Furthermore, we leverage tools like Argo CD for continuous integration and delivery (CI/CD), enabling us to deliver updates and improvements to our infrastructure seamlessly and regularly. This constant evolution of our infrastructure, driven by user needs and feedback, ensures we stay at the cutting edge of data science technology and best practices.

Our emphasis on user interaction and collaboration not only fosters a dynamic and innovative research environment but also keeps us grounded in the needs of our users, ensuring our infrastructure is both powerful and accessible.

VI. USE CASES

In the two years since the DSRS's inception, several significant research projects have been carried out using its infrastructure as well as some general tasks as shown in Figure 2. Among these, we include research on tax and information disclosure from thousands of companies across the globe. Another example is a large-scale study of consumer credit data panels in the US. These projects not only showcased the computational prowess of DSRS but also underscored its versatility in catering to diverse business research needs. A typical example would look like this:

A faculty member at the Gies College of Business, is leading a team of researchers to understand the potential market dynamics for a new sustainable product launch targeting millennials and Gen Z consumers. To gather real-time data from social media platforms, eCommerce websites, and sustainability forums, process it, and derive actionable insights for the niche product launch.

Implementation with DSRS:

Data Collection: Using the DSRS platform, the team utilized web scraping tools integrated with JupyterHub to gather data from targeted sources.

Data Processing: Through the DSRS's advanced data processing services, raw data was cleaned, transformed, and structured. This was facilitated by the platform's intuitive interfaces and seamless access to computational resources.

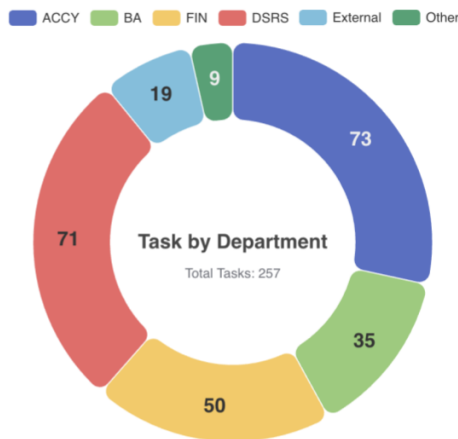


Fig. 2. Tasks summary across different departments, Accountancy (ACCY), Business Administration (BA), Finance (FIN), and others

Analysis: Leveraging DSRS’s machine learning and natural language processing capabilities, the team was able to identify prevailing sentiments, purchase intentions, and the most discussed product features among the target audience.

Visualization: DSRS’s integration with visualization tools enabled the faculty’s team to create real-time dashboards, highlighting market sentiment shifts and potential demand pockets.

VII. CONCLUSION AND FUTURE DIRECTION

The Data Science Research Services (DSRS) at Gies College of Business has proven to be an invaluable asset for data-driven research in our community. Our comprehensive infrastructure provides a diverse set of tools and applications that empower researchers to answer complex business questions. By seamlessly integrating data management, computation, and collaboration, we are enabling the Gies College of Business to be at the forefront of technological advancements and data capability.

Our efforts have built a strong foundation, but the dynamic nature of data science calls for continued evolution. Looking ahead, our aim is to continuously expand our services, staying abreast of emerging technologies, trends, and user needs. We are particularly excited about advancements in the field of artificial intelligence, and machine learning and are looking to incorporate more of these capabilities into our service offerings. We already started incorporating LLM’s to our suite of tools for researchers to use during their studies.

Furthermore, we are exploring opportunities to increase our interconnections across the wider academic and industry communities. For instance, we are examining potential collaborations with other academic units within the University and industry partners, with the goal of enriching our research landscape and providing real-world applications for our data science efforts.

Lastly, as a core part of our mission, we are committed to fostering a nurturing environment for the next generation of data scientists. We will continue to provide opportunities

for students to work on live projects, allowing them to gain valuable hands-on experience in data science and machine learning.

The journey of the DSRS is one of continuous growth and learning. We are excited about the opportunities ahead and remain dedicated to our mission of making the Gies College of Business the most technologically forward and data-capable business college in the world.

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